

Big Data Analytics on App Store Rating Patterns

Do Free Apps Get Rated Differently Than Paid Apps?

Module: 6CS030 Big Data

Topic Type: Data Analysis

Context

- Google Play Store hosts over 2.6 million apps free and paid
- Users rely on star ratings and reviews before downloading
- Ratings directly influence app visibility, downloads, and developer revenue
- Rating data is a genuine Big Data problem millions of reviews daily
- Traditional analysis cannot handle this scale ML techniques are needed

100B+

App downloads on Google Play

Big Data Dimensions:

Volume · Variety · Velocity

Research Question

Does pricing model (free vs. paid) influence how users rate mobile applications?

Why Big Data Techniques Are Needed

Volume

Millions of app records and reviews — too large for manual analysis

Variety

Structured metadata (price, installs) + unstructured text reviews combined

ML Need

Pattern detection across categories requires classification models (sklearn)

Scale

Cross-platform comparison (Google Play + Apple App Store) amplifies data size

AIM

To analyse app store rating patterns across free and paid applications using big data analytics and ML techniques on Google Play and Apple App Store datasets.

Objectives

- 1 Collect and preprocess large-scale app metadata from Google Play and Apple App Store (GitHub datasets)
- 2 Perform exploratory data analysis (EDA) comparing free vs. paid app ratings across all categories
- 3 Identify correlations between pricing, installs, review count, and user ratings
- 4 Build an ML classifier (Random Forest / Logistic Regression) to predict whether an app is free or paid
- 5 Visualise and communicate findings through charts, heatmaps, and structured report

Existing research is grouped into three main categories:

01

App Review Mining & Analysis

Studying what users say in reviews — requirements elicitation, bug detection, user satisfaction

02

Rating Prediction Using ML

Predicting app ratings from metadata features using ensemble and classical ML classifiers

03

Sentiment Analysis of Reviews

Classifying review tone (positive / negative / neutral) using NLP and deep learning methods

Our work sits at the intersection of all three — combining rating pattern analysis, pricing comparison, and ML classification

[1] Hasnain et al., ACM, 2023

User Reviews of Top Mobile Apps in Apple and Google App Stores

Finding: Studied user feedback mechanisms on both platforms. Reviews serve as valuable data beyond bug reporting — enabling requirements elicitation and developer improvement.

Gap: Focused only on top-ranked apps; free vs. paid pricing not examined.

[2] Kalampokis et al., Springer, 2022

Analysing App Reviews for Software Engineering: A Systematic Literature Review

Finding: Reviewed 126 studies covering requirements engineering, bug detection, and user satisfaction analysis using app store reviews.

Gap: Pricing model not used as a variable; no ML classification performed.

[3] Stocchi et al., Springer, 2022

Marketing Research on Mobile Apps: Past, Present and Future

Finding: Synthesised 471 studies showing how app ratings shape customer experience and competitive advantage in mobile marketing.

Gap: Marketing perspective only — no big data or ML analytical approach applied.

[4] Wieringa et al., Springer, 2022

Combinatory Role of Online Ratings and Reviews in Mobile App Downloads

Finding: Tracked 341 apps for nearly 2 years. Rating score combined with review sentiment significantly impacts download decisions.

Gap: Apple App Store only; no Google Play or free vs. paid comparison performed.

[5] IEEE ICITEICS, 2024

Predicting App Ratings on Google Play Store: An Ensemble Learning Approach

Finding: Used ensemble ML models (Random Forest, Gradient Boosting) to predict app ratings with strong accuracy from app metadata.

Gap: Focused on rating prediction only — pricing model not used as key analysis variable.

[6] Qureshi et al., Springer, 2023

Performance Evaluation of ML Models on Large Dataset of Android App Reviews

Finding: Evaluated multiple ML classifiers on Play Store review data. Ensemble methods outperform single classifiers for review classification.

Gap: Sentiment-focused study — did not analyse rating differences between free and paid apps.

[7] Elsevier, 2024

Sentiment Analysis on E-Commerce Product Reviews Using ML and DL: A Systematic Literature Review

Finding: Reviewed 20 studies on ML/DL sentiment classification. LSTM and CNN-based models achieve highest accuracy — up to 98% F1-score.

Gap: E-commerce focused; does not address app store pricing dynamics or free vs. paid patterns.

[8] Springer, 2025

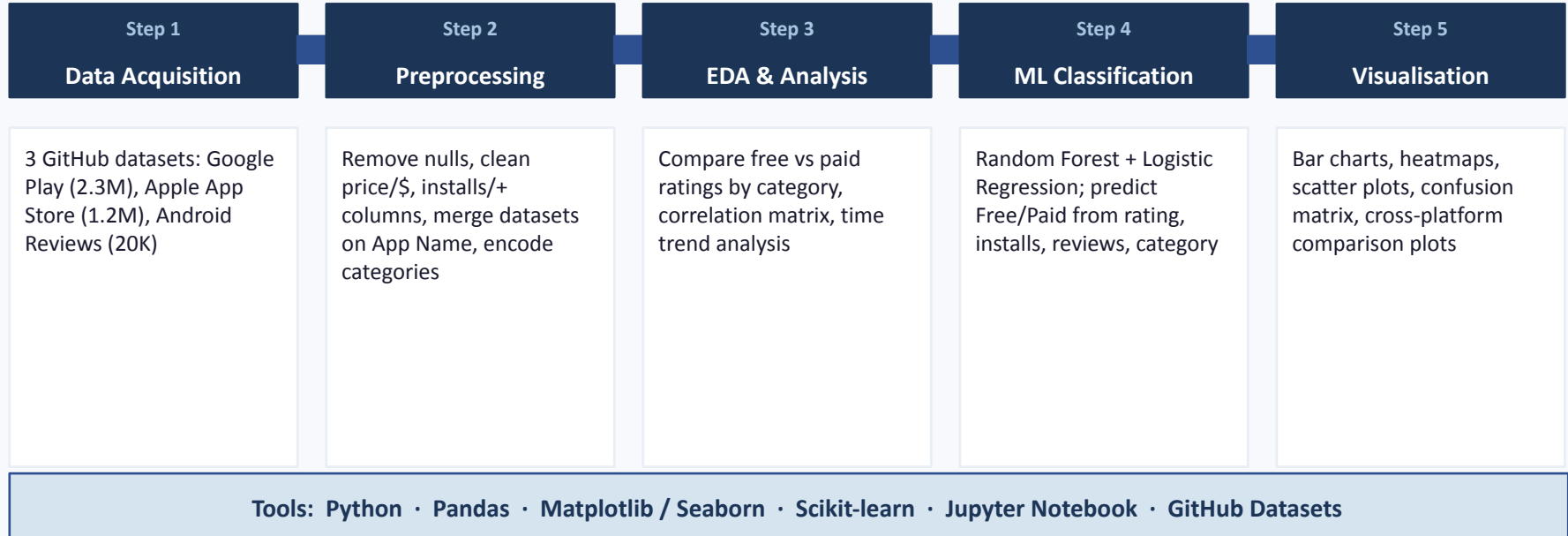
Investigating User Reviews of Diverse Age Groups' Mobile Apps Using ML

Finding: Analysed mobile app reviews across children, teen, and adult categories using Naïve Bayes classifier and BERT-LDA hybrid topic modelling.

Gap: Age-group focused — pricing model (free vs. paid) not investigated as a factor.

Summary — What Is Missing Across All 8 Papers

Existing Work	What It Covered	Gap Identified
App review mining	Review content, bugs	No free vs. paid comparison
Rating prediction (ML)	Predicting star ratings	Pricing not used as variable
Sentiment analysis	Positive/negative tone	No cross-platform pricing study



Datasets

Dataset	Size	Source (GitHub)
Google Play Store	2.3M apps, 24 attributes	github.com/gauthamp10/Google-Playstore-Dataset
Apple App Store	1.2M apps, 21 attributes	github.com/gauthamp10/apple-appstore-apps
Android App Reviews	20K labelled reviews	github.com/amitt001/Android-App-Reviews-Dataset

[1] Hasnain M. et al., "User Reviews of Top Mobile Apps in Apple and Google App Stores," Communications of the ACM, 2023.

[2] Kalampokis E. et al., "Analysing App Reviews for Software Engineering: A Systematic Literature Review," Empirical Software Engineering, Springer, 2022.

[3] Stocchi L. et al., "Marketing Research on Mobile Apps: Past, Present and Future," Journal of the Academy of Marketing Science, Springer, 2022.

[4] Wieringa J. et al., "The Combinatory Role of Online Ratings and Reviews in Mobile App Downloads," Journal of Marketing Analytics, Springer, 2022.

[5] "Predicting App Ratings on Google Play Store: An Ensemble Learning Approach," IEEE ICITEICS, 2024.

[6] Qureshi A. et al., "Performance Evaluation of ML Models on Large Dataset of Android App Reviews," Multimedia Tools and Applications, Springer, 2023.

[7] "Sentiment Analysis on E-Commerce Product Reviews Using ML and DL: A Systematic Literature Review," ScienceDirect, Elsevier, 2024.

[8] "Investigating User Reviews of Diverse Age Groups' Mobile Apps Using ML," Quality & Quantity, Springer, 2025.